

Empirical Proof Record: VALIDATED

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Proof ID: 779d65190263daa39fcd53f54437a5f1...

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1. Claim

When an identically-recognised probe is re-exposed across a trajectory, it lands on a strictly-deeper (more-scarred) manifold every time: $\text{memory_path_dependence} = (\text{recognised re-exposure transitions with } \Delta \text{ permanent-scars} > 0) / (\text{recognised re-exposure transitions}) \geq 1.0$, with the whole-run scar count monotonic non-decreasing (a scar cannot be reset). This measures memory where Kimera keeps it — the scars and the S4 manifold — not the content-recognition layer.

- **Operationalisation:** Run 8 stimuli each re-exposed 3x, interleaved ($\text{gap} = \text{len}(\text{stimuli})$), through Kimera's entity target (Takwin) in batch mode so the substrate accumulates across the batch. Per cycle read `vault_stats.total_scars_stored` (canonical permanent scars), `arachne_web_coupling_frobenius` (manifold deformation), and the concept set (recognition). For each same-stimulus consecutive exposure pair: recognised iff $\text{Jaccard}(\text{concepts}) \geq 0.80$; confirmed iff recognised AND $\Delta \text{ scars} > 0$. $\text{memory_path_dependence} = \text{confirmed} / \text{recognised}$. Forced to 0.0 (REFUTED) if the whole-run scar count ever decreases.
- **Threshold:** $\text{memory_path_dependence} \geq 1.0$ proportion
- **H0:** H0: $\text{memory_path_dependence} < 1.0$ — some re-exposure of a recognised probe did NOT land on more permanent scars (the substrate re-derives without accumulating), or a scar was reset; not path-dependent permanent memory
- **H1:** H1: $\text{memory_path_dependence} \geq 1.0$ — every re-exposure of a recognised probe lands on a strictly-deeper manifold and no scar is ever reset; path-dependent permanent memory in the scar/S4 substrate

2. Verdict

VALIDATED — observed 1.0
memory_path_dependence 1.0000 = 16/16 recognised re-exposure transitions landed on strictly more permanent scars; 0 transitions failed recognition (excluded). Permanence: HOLDS (0 scar-count decreases over the run). Scars 1.0→24.0; coupling 1.337322→11.262303; phi |delta| mean 0.0000 (Φ -rigid → memory is in the scars/manifold, not Φ); scar-depth vs exposure-ordinal Spearman rho=0.9436, p=2.4e-12 (deepens with experience)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
0.flow.memory_path_dependence	memory_path_dependence	1.0	—	ophamin 0.115.2

4. Pre-registration

- Registered at: 2026-05-23T23:12:01.650531+00:00
- Config hash: f5c0c3ae55bb8ed69287755ab991315a...
- Data hash: 4e6090352a40ea27df1cf08fb4d02bd5...

5. Signature

67b0dcfb80cc510b0f46f5067fb112e2625db86146434b453f517726e154bf3d